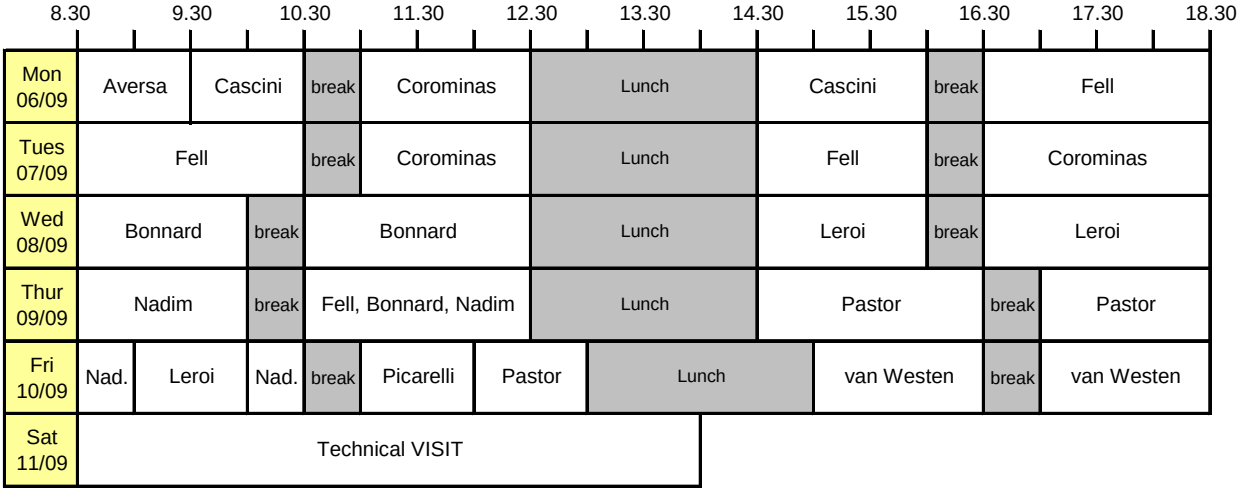
# Programme of the LARAM School 2010: 1st week




Special Session [1.0 hours]

**Aversa**: Risk categories within the Euro Codes

SESSION [4.0 hours] "Introduction to landslide analysis"

**Cascini** (1.0 hours): Introduction to landslides.

**Corominas** (1.5 hours): Landslides typology and classification. Geological analyses.

**Cascini** (1.5 hours): The geotechnical slope model.

SESSION [23.0 hours] "Risk Theory and Risk Analysis for Landslides"

**Fell** (2.0 hours): Landslide Risk Management concepts and framework.

**Fell** (2.0 hours): Landslide Frequency Assessment.

**Corominas** (1.5 hours): Empirical and semi-empirical models for run-out and travel distance.

**Fell** (1.5 hours): Application of QRA to crater lake hazard and case study of debris flow hazard and risk analysis.

**Corominas** (2.0 hours): Case studies rockfall hazard and risk analyses.

**Bonnard** (1.5 hours): Different components of vulnerability to landslides. Case examples?.

**Bonnard** (2.0 hours): Long-term efficiency of protection works and residual risks.

**Leroi** (1.5 hours): Tolerable risk criteria and the practical management of risk, case example?

**Leroi** (2.0 hours): Landslide Risk Management at La Desirade Island and Pointe-Noir in Guadeloupe.

**Nadim** (1.5 hours): Deterministic and probabilistic models for slope stability evaluation.

**Fell, Bonnard, Nadim** (1.5 hours): Case Studies: coal waste dump risk assessment (Fell), example from motorway in La Reunion Island ( Bonnard), Aknes Rock slope in Norway ( Nadim).

**Pastor** (2.0 hours): Introduction to modelling of catastrophic landslide events.

**Pastor** (1.5 hours): Advanced numerical models: initiation of landslides, propagation of sediments.

Special Session [7.0 hours] "The EU FP7 SafeLand project"

**Nadim** (0.5 hours): Introduction to SafeLand Project.

**Leroi** (1.0 hours): Demography and its influence on landslide hazard. Who are the stakeholders in a landslide risk management process?

**Nadim** (0.5 hours): Human-induced landslides.

**Picarelli** (1.0 hours): SafeLand Area 1: Triggering factors for different types of landslides. Summary of other results from Area 1.

**Pastor** (1.0 hours): SafeLand Area 1: Models and methods for estimating runout distance.

**Vav Westen** (3.0 hours): Use of GIS for landslide risk assessment.

Technical Visit [5.0 hours]

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# Programme of the LARAM School 2010: 2nd week

| | 8.30 | 9.30 | 10.30 | 11.30 | 12.30 | 13.30 | 14.30 | 15.30 | 16.30 | 17.30 | 18.30 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Mon 13/09 | Cascini | | break | Cascini | Calvello | Lunch | | Calvello | break | Godt | |
| Tue 14/09 | Godt | | break | Godt | Sorbino | Lunch | | Pang | break | Pang | |
| Wed 15/09 | Pang | | break | Cascini | Scavia | Lunch | | Scavia | break | Alonso | |
| Thur 16/09 | Alonso | | break | Darve | | Lunch | | (Mem Var) | break | Sorbino | |
| Fri 17/09 | Lacerda | | break | Lacerda | | Lunch | | Ferlisi | break | Ferlisi | Cascini, Lacerda |
| Sat 18/09 | Departures | | | | | | | | | | |

SESSION [15.5 hours] “Landslide susceptibility, hazard and risk zoning: international experiences”

**Cascini** (2.0 hours): Input elements to zoning maps. Zoning scales, levels and methods.

**Cascini** (1.0 hours): Basic methods and procedures for zoning at small and medium scales (<1:100,000 – 1:25,000).

**Calvello** (1.0 hours): Tutorial on empirical methods for susceptibility, hazard and risk zoning at 1:25,000 scale.

**Calvello** (1.0 hours): Statistical methods for susceptibility and hazard analyses.

**Godt** (2.0 hours): Landslide hazard and risk zoning at small scale: the USGS experience. Physics of infiltration and shallow landslide initiation: the TRIGRS model.

**Godt** (2.5 hours): Tutorial on the TRIGRS model.

**Sorbino** (1.5 hours): Susceptibility analysis of shallow landslides source areas using physically based models: the Pizzo d’Alvano case study.

**Pang** (1.5 hours): Landslide risk zoning at large scale: the GEO experience (I).

**Pang** (1.5 hours): Landslide risk zoning at large scale: the GEO experience (II).

**Pang** (1.5 hours): Tutorial on QRA at large scale.

SESSION [9.5 hours] “The role of sophisticated methods in landslide Risk analysis”

**Cascini** (1.0 hours): Introduction to advanced slope stability characterization.

**Scavia** (1.0 hours): Fracture mechanics approach for the analysis of progressive failure in slopes (part I)

**Scavia** (1.5 hours): Fracture mechanics approach for the analysis of progressive failure in slopes (part II)

**Alonso** (1.0 hours): Rapid drawdown and risks associated.

**Alonso** (1.5 hours): Fundamentals of thermo-hydraulic-mechanical interactions in soils.

**Alonso** (1.0 hours): Rapid landslides. Practical criteria.

**Darve** (2.5 hours): Geomechanics applied to landslide modelling.

Special Session [1.0 hours] “Memorial for Ioannis Vardoulakis”

**Cascini, Papamichos**

SESSION [7.0 hours] “Landslide risk mitigation”

**Sorbino** (2.0 hours): The role of control works in the risk mitigation framework. Principles and design of control works against fast landslides.

**Lacerda** (1.5 hours): Site investigation and field monitoring in the research of sliding mechanisms of residual and colluvial slopes in tropical areas.

**Lacerda** (2.0 hours): Principles and design of control works against rockfalls and shallow slides: solution in urban areas - the example of Rio de Janeiro.

**Ferlisi** (2.0 hours): Physical vulnerability to slow-moving landslides.

CONCLUDING REMARKS [1 hour]

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